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Team Name : ByteRank AI

Team Leader Name : Mohammed Abdul Wasey

Problem Statement :

Developing an Explainable AI-Powered Candidate Ranking System that ranks 100,000 candidate profiles using hybrid scoring, semantic search, and recruiter-friendly reasoning.

Solution Overview

- **What is your proposed solution?**
 - Explainable AI-Powered Candidate Ranking System
 - Ranks 100,000 candidate profiles against a job description
 - Uses semantic search beyond keyword matching
 - Combines semantic, behavioural, consistency & recruiter signals
 - Produces Top 100 candidates with explainable reasoning
- **What differentiates your approach from traditional candidate matching systems?**
 - Evaluates complete candidate profiles instead of keywords alone
 - Hybrid scoring improves ranking accuracy and reliability
 - Generates transparent reasoning for every shortlisted candidate
 - Reduces black-box decision making through Explainable AI
 - Scalable pipeline for 100,000 candidate profiles

JD Understanding & Candidate Evaluation

- **What are the key requirements extracted from the JD?**
 - Extracts required skills, role and experience level
 - Identifies domain knowledge and job responsibilities
 - Detects retrieval technologies like FAISS, BM25 and RAG
 - Builds a semantic representation of the job description
 - Creates structured features for accurate candidate matching
- **Which candidate signals are most important for determining relevance? / How does your solution evaluate candidate fit beyond keyword matching?**
 - Measures semantic similarity between candidate and job description
 - Uses behaviour and consistency scores for profile quality
 - Incorporates recruiter response and GitHub activity signals
 - Evaluates experience, job title and skill alignment
 - Combines all signals through explainable hybrid scoring

Ranking Methodology

- **How does your system retrieve, score, and rank candidates?**
 - Retrieves candidates using semantic search
 - Scores semantic, behavioural and consistency signals
 - Applies recruiter and profile quality signals
 - Produces an explainable Top 100 ranking
- **What models, algorithms, or heuristics are used?**
 - Sentence Transformers for semantic embeddings
 - Cosine similarity for candidate retrieval
 - Weighted hybrid scoring for ranking
 - Rule-based explainable reasoning generation
- **How are multiple candidate signals combined into a final ranking?**
 - Normalizes all candidate scores
 - Applies weighted hybrid score aggregation
 - Ranks candidates by final composite score
 - Generates transparent recruiter-friendly explanations

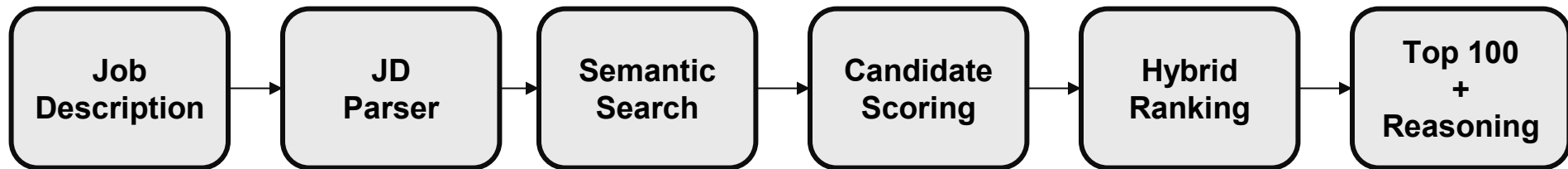
Explainability & Data Validation

- **How are ranking decisions explained?**
 - Explains every ranking with matched candidate signals
 - Highlights relevant skills and experience
 - Justifies recruiter and profile quality scores
 - Provides transparent recruiter-friendly reasoning
- **How do you prevent hallucinations or unsupported justifications?**
 - Uses only verified profile information
 - Generates reasoning from computed scores
 - Avoids unsupported or fabricated claims
 - Ensures consistent and explainable outputs
- **How does your solution handle inconsistent, low-quality, or suspicious profiles?**
 - Penalizes inconsistent profile information
 - Uses behaviour and consistency validation
 - Reduces scores for weak profile quality
 - Prioritizes reliable candidate profiles

End-to-End Workflow

- **What is the complete workflow from JD input to ranked candidate output?**
 - Job Description (JD) is parsed and key requirements are extracted
 - Candidate profiles are converted into structured representations
 - Semantic search retrieves the most relevant candidates
 - Behaviour, consistency and recruiter signals are evaluated
 - Hybrid weighted scoring computes the final candidate score
 - Candidates are ranked and Top 100 are shortlisted
 - Explainable reasoning is generated for every shortlisted candidate
 - Final ranked list with scores and explanations is returned

System Architecture



Results & Performance

- **What results or insights demonstrate ranking quality?**
 - Accurately ranks candidates using semantic relevance
 - Combines multiple signals for reliable ranking
 - Generates transparent recruiter-friendly reasoning
 - Produces a high-quality Top 100 candidate shortlist
- **How does your solution meet the challenge's runtime and compute constraints?**
 - Uses FAISS for fast candidate retrieval
 - Scores only the most relevant candidates
 - Lightweight hybrid scoring reduces computation
 - Efficiently scales to 100,000 candidate profiles

Technologies Used

- **What technologies, frameworks, and tools were used and why were they selected for this solution?**
 - Python – Core application development
 - Sentence Transformers – Semantic candidate matching
 - FAISS – Fast vector similarity search
 - Pandas & NumPy – Data preprocessing
 - JSON – Candidate and JD data handling
 - Hybrid scoring – Explainable candidate ranking

Submission Assets

- GitHub Repository
<https://github.com/Abdul-Wasey-M/redrob-ai-ranking>
- Project Presentation (PPT)
- Final Submission File
outputs/final_submission.csv
- Project Documentation
README.md
- Demo Video (GitHub Repository Walkthrough)



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THANK YOU